

HW - 7

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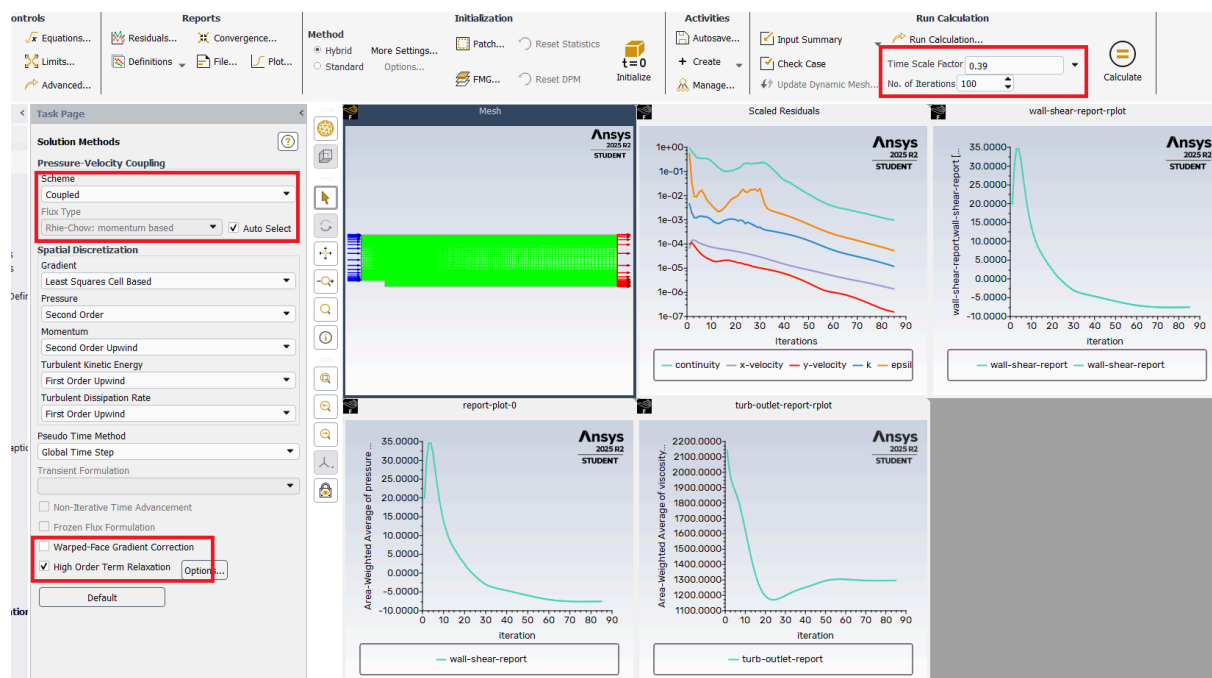
IISC - CFD

Convergence plot with timescale factor 0.39 (0.4 oscillates)

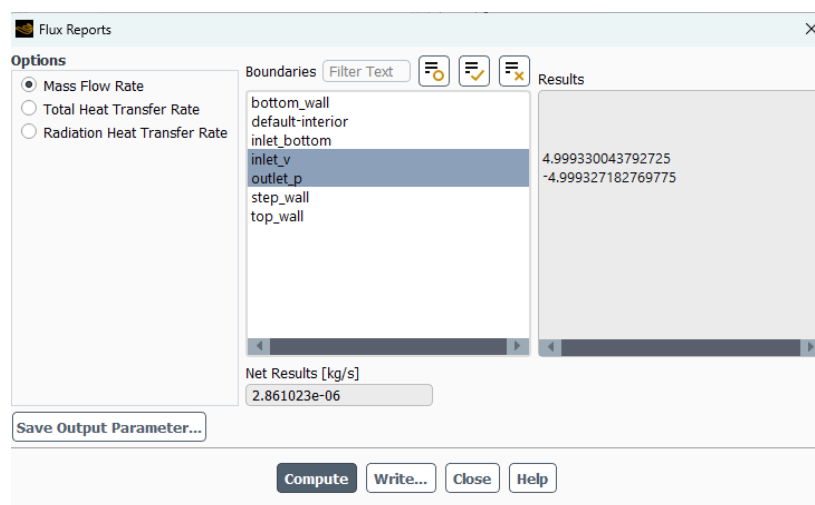
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iter  continuity  x-velocity  y-velocity  k  epsilon  time/iter
78  1.3915e-03  2.0397e-06  2.5708e-07  1.9129e-05  8.5030e-05  0:00:03  22
79  1.3163e-03  1.9342e-06  2.3441e-07  1.7957e-05  7.9897e-05  0:00:03  21
80  1.2349e-03  1.8327e-06  2.1468e-07  1.6831e-05  7.4976e-05  0:00:03  20
81  1.1690e-03  1.7355e-06  1.9768e-07  1.5754e-05  7.0241e-05  0:00:03  19
82  1.1120e-03  1.6427e-06  1.8373e-07  1.4728e-05  6.5683e-05  0:00:03  18
83  1.0549e-03  1.5546e-06  1.7233e-07  1.3749e-05  6.1336e-05  0:00:03  17
84  1.0149e-03  1.4705e-06  1.6271e-07  1.2824e-05  5.7192e-05  0:00:02  16
85  9.7906e-04  1.3908e-06  1.5429e-07  1.1946e-05  5.3268e-05  0:00:02  15
! 85 solution is converged
Registering ReportDefFiles, (" \turb-outlet-report-rfile.out")
Writing "I gzip -2cf > SolutionMonitor.gz"...
Writing temporary file C: \Temp\flntgz-357215 ...
Done.

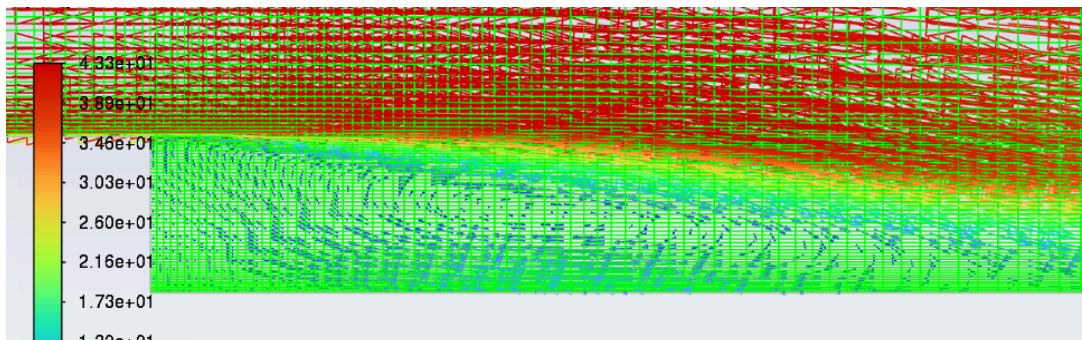
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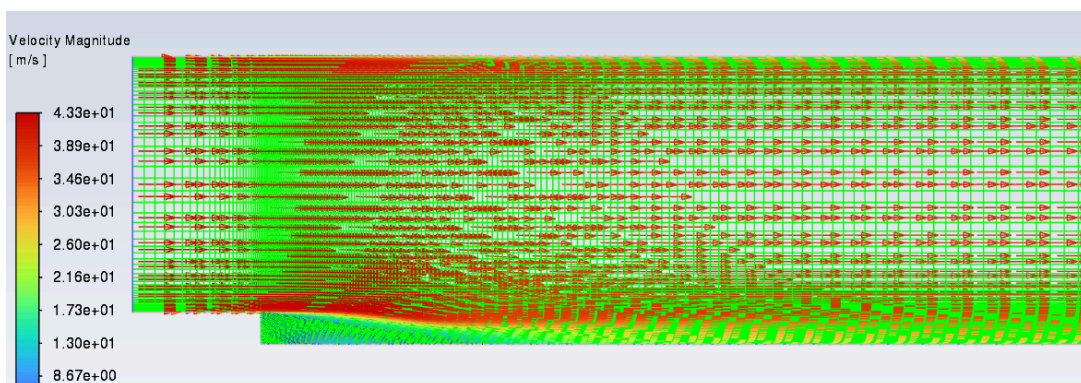
Mass flow rate of inlet and outlet. The value is close to 0 as expected.



Vectors showing re-circulation zones are as below.

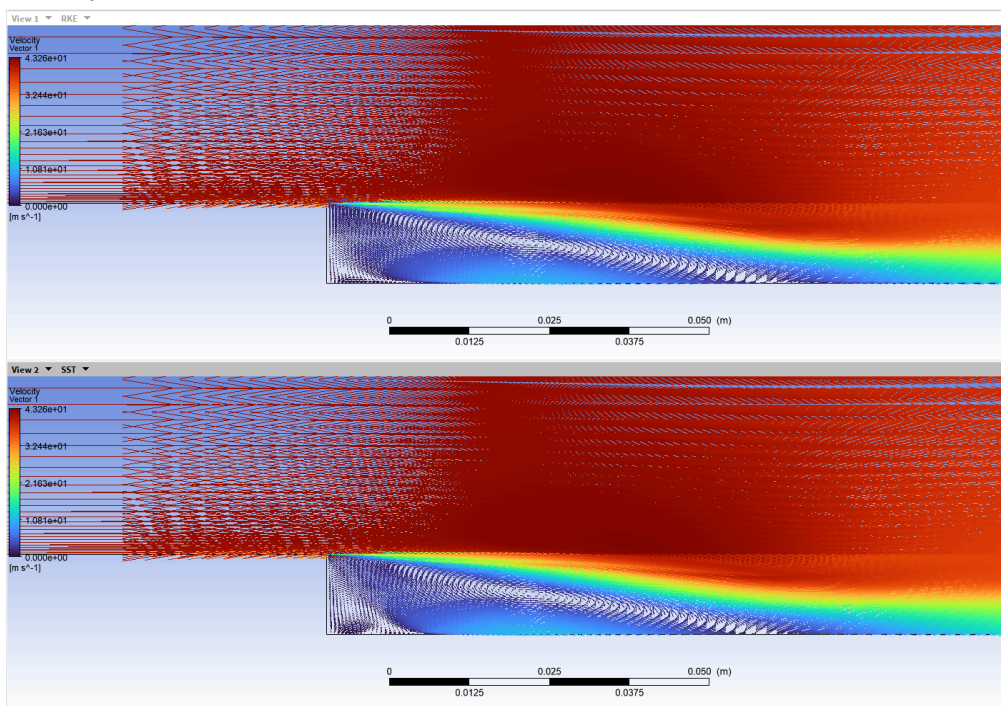


Velocity magnitude

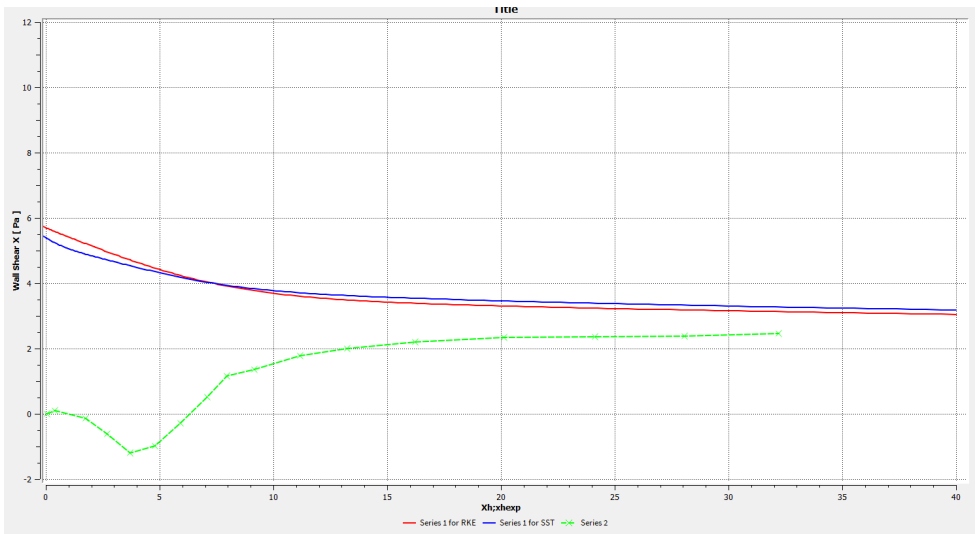


Comparison between RKE and SST

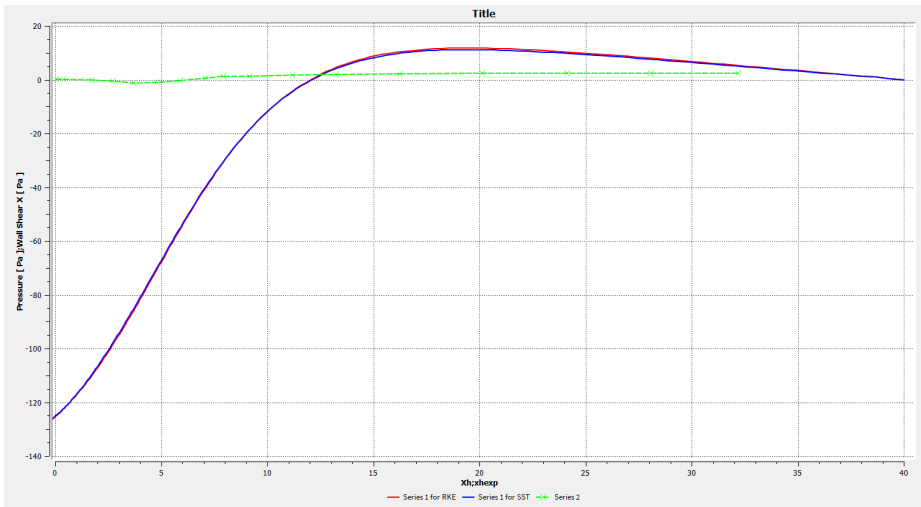
Velocity vector plots



Plot for x-wall-shear-ds (bottom_wall) and top wall sheer with xh step expression.



The wall shear stress on the upper wall is positive throughout and decreases along the flow direction. It drops rapidly near the step due to flow expansion and then gradually levels off as the boundary layer develops downstream. This behavior is physically consistent, since there is no flow separation on the top wall and the velocity gradient at the wall reduces as the flow progresses.



The pressure distribution shows a rapid recovery downstream of the step due to flow expansion, followed by a gradual decrease due to wall friction, which is consistent with expected backward-facing step flow behavior.

Production of k, contour lines

RKE



SST



The contours of turbulence kinetic energy production show high values immediately downstream of the step, concentrated along the shear layer between the main flow and the recirculation region. The production is strongest near the step edge and gradually decreases downstream. This location is expected, as turbulence production is highest in regions with strong velocity gradients, which occur in the separated shear layer formed after the step